TI-99/4A — Chip Pinouts

MAJOR SOCKETED ICS · TMS9900 · TMS9918A · TMS9919 · TMS9901 · RAM · TEST & REPAIR

DIP pinouts seen from the top, notch up, pin 1 at top-left. See the **Legend** card for notation & colour key. The TMS9900 needs **−5 / +5 / +12V**.

Legend — reading the pinouts
notation

`/NAME` **Active LOW** — true / asserted when the pin is at logic **LOW** (~0 V). A leading **/** (datasheets often use an overline) marks these: e.g. `/RES /IRQ /CS /WE`.

`NAME` **Active HIGH / normal** — asserted when the pin is **HIGH** (~+5 V). No slash. `R/W` is a level: HIGH = read, LOW = write.

Colour key: Vcc / Vdd = power · **GND / Vss = ground** · **φ = clock**. **Orientation:** chip seen from the top, notch up — pin 1 top-left, numbers run **down the left** then **up the right**.

TMS9900 16-bit CPU
64-DIP

Vbb -5V	1	64	/HOLD
Vcc +5V	2	63	/MEMEN
WAIT	3	62	READY
/LOAD	4	61	/WE
HOLDA	5	60	CRUCLK
/RESET	6	59	Vcc +5V
IAQ	7	58	NC
φ1	8	57	NC
φ2	9	56	D15
A14	10	55	D14
A13	11	54	D13
A12	12	53	D12
A11	13	52	D11
A10	14	51	D10
A9	15	50	D9
A8	16	49	D8
A7	17	48	D7
A6	18	47	D6
A5	19	46	D5
A4	20	45	D4
A3	21	44	D3
A2	22	43	D2
A1	23	42	D1
A0	24	41	D0
φ4	25	40	Vss
Vss	26	39	NC
Vdd +12V	27	38	NC
φ3	28	37	NC
DBIN	29	36	IC0
CRUOUT	30	35	IC1
CRUIN	31	34	IC2
/INTREQ	32	33	IC3

16-bit CPU @ 3 MHz. Needs THREE rails — **−5V** (Vbb), **+5V** (Vcc 2&59), **+12V** (Vdd) — and a 4-phase clock φ1–φ4 from the TIM9904. **Fails:** dead console, no title screen. Check all three voltages first.

TMS9918A VDP video
40-DIP

/RAS	1	40	XTAL2
/CAS	2	39	XTAL1
AD7	3	38	CPUCLK
AD6	4	37	GROMCLK
AD5	5	36	COMVID
AD4	6	35	EXTVDP
AD3	7	34	/RESET
AD2	8	33	Vcc
AD1	9	32	RD0
AD0	10	31	RD1
R/W	11	30	RD2
Vss	12	29	RD3
MODE	13	28	RD4
/CSW	14	27	RD5
/CSR	15	26	RD6
/INT	16	25	RD7
CD7	17	24	CD0
CD6	18	23	CD1
CD5	19	22	CD2
CD4	20	21	CD3

Video processor with its own 16KB VRAM (4116). COMVID composite out. **Fails:** no / garbled picture, bad colour.

TMS9919 / SN94624 PSG sound
16-DIP

D5	1	16	Vcc
D6	2	15	D4
D7	3	14	CLOCK
/READY	4	13	D3
/WE	5	12	D2
/CE	6	11	D1
AUDIO OUT	7	10	D0
GND	8	9	AUDIO IN

Sound (SN76489 family). On the TI, **pin 9 = AUDIO IN** (speech / cassette mix) where a plain SN76489 has NC. **Fails:** no / distorted sound.

TMS9901 PSI I/O
40-DIP

/RST1	1	40	Vcc
CRUOUT	2	39	S0
CRUCLK	3	38	P0
CRUIN	4	37	P1
/CE	5	36	S1
/INT6	6	35	S2
/INT5	7	34	P15
/INT4	8	33	P14
/INT3	9	32	P13
φ	10	31	P12
/INTREQ	11	30	P11
IC3	12	29	P10
IC2	13	28	P9
IC1	14	27	P8
IC0	15	26	P2
Vss	16	25	S3
/INT1	17	24	S4
/INT2	18	23	P7
P6	19	22	P3
P5	20	21	P4

Programmable Systems Interface: keyboard scan, interrupts, cassette control & timer over the CRU serial bus. **Fails:** dead keyboard, no cassette, stuck interrupts.

Memory & GROM
RAM/ROM

VRAM	16KB — 8× 4116 (VDP)
Scratchpad	256B fast static RAM
Console ROM	12KB — GROM + ROM
GROM	serial Graphics ROM — odd 1-wire bus

Fails: garbled graphics (VRAM), won't boot (GROM/ROM). GROMs are a TI quirk.

Test & Repair
tips

Rails	−5V, +5V, +12V all required
Clock	TIM9904 4-phase φ
No title	TMS9900 / clock / ROM
No picture	9918A + 4116 VRAM
Dead keys	TMS9901
No sound	TMS9919

The 99/4A is unusual: 16-bit CPU, GROMs, and three supply rails. Verify voltages before swapping the big 64-pin TMS9900.